

Please check the examination details below before entering your candidate information	
Candidate surname	Other names
Centre Number	Candidate Number
Pearson Edexcel Level 3 GCE	
Monday 23 June 2025	
Afternoon (Time: 1 hour 30 minutes)	Paper reference 9FM0/4B
Further Mathematics Advanced PAPER 4B: Further Statistics 2	
You must have: Mathematical Formulae and Statistical Tables (Green), calculator	Total Marks

Candidates may use any calculator permitted by Pearson regulations. Calculators must not have the facility for symbolic algebra manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions

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Answers without working may not gain full credit.
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Information

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- There are 7 questions in this question paper. The total mark for this paper is 75.
- The marks for **each** question are shown in brackets
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Advice

- Read each question carefully before you start to answer it.
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Turn over ►

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1. A medical researcher is exploring the relationship between the weight, x kg, and the head circumference, y cm, of newborn babies. A random sample of 15 newborn babies is taken and the data are summarised by the following statistics

$$\sum x = 50.46 \quad \sum x^2 = 171.828 \quad \sum y = 518.9 \quad \sum y^2 = 18004.47 \quad S_{xy} = 7.8284$$

- (a) Show that, to 3 decimal places, $S_{xx} = 2.081$ and $S_{yy} = 53.989$ (3)

- (b) Find the equation of the regression line of y on x , giving your answer in the form $y = a + bx$ (3)

- (c) Find the residual sum of squares (RSS) for these data. (1)

One of these 15 babies had a birth weight of 3.26 kg and a head circumference of 36.8 cm

- (d) Find the residual for this baby. (2)

The researcher claims that this baby could be an outlier.

- (e) Using your answers to part (c) and part (d), give a reason that might support this claim. (1)



DO NOT WRITE IN THIS AREA

Question 1 continued

Lined area for writing the answer to Question 1.



Question 1 continued

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Question 1 continued

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(Total for Question 1 is 10 marks)



2. Two lecturers are marking a large batch of students' essays, to place them in rank order.

A random sample of 9 students is taken and their essays are marked by both lecturers.

The marks awarded are given in the table below.

Student	<i>A</i>	<i>B</i>	<i>C</i>	<i>D</i>	<i>E</i>	<i>F</i>	<i>G</i>	<i>H</i>	<i>I</i>
Lecturer 1	10	16	5	17	12	7	14	11	8
Lecturer 2	17	22	19	24	21	15	23	18	20

- (a) Calculate Spearman's rank correlation coefficient for these data. (4)
- (b) Carry out a suitable test to assess whether the lecturers agree about the rank order of the essays. You should use a 5% level of significance and state your hypotheses and critical value clearly. (3)



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Question 2 continued

Lined area for writing answers.

(Total for Question 2 is 7 marks)



3. A scientist investigated the effects of nitrogen-fixing bacteria on cereal crop yield, $x \text{ kg ha}^{-1}$

The data come from independent random samples from normal distributions.

The results are summarised in the table below.

	n	$\bar{x}$	s_x
Without nitrogen-fixing bacteria	6	1648	143.8
With nitrogen-fixing bacteria	5	1910	64.0

- (a) Test whether or not the variances of the cereal crop yields are the same.
You should use a 10% level of significance and state your hypotheses and critical value clearly.

(4)

A suitable test to determine whether there is an increase in mean cereal crop yield has a p -value of 0.002

- (b) (i) Explain the importance of the conclusion of the test in part (a) to this test.
(ii) Comment on the effectiveness of nitrogen-fixing bacteria on cereal crop yields.

(2)



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Question 3 continued

Lined area for writing the answer to Question 3.

(Total for Question 3 is 6 marks)



DO NOT WRITE IN THIS AREA

Question 4 continued

Lined area for writing the answer to Question 4.



Question 4 continued

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Question 4 continued

Lined area for writing answers.

(Total for Question 4 is 13 marks)



5. The random variable $X \sim U[1, 4]$

(a) Find

(i) $P(1.8 < X < 3.2)$

(ii) $P(X > 3.2 \mid X > 1.8)$

(3)

A random sample of 10 observations of X is taken.

The random variable M represents the maximum value of these 10 observations.

The cumulative distribution function of M , $F(y)$, is given by

$$F(y) = \begin{cases} 0 & y < 1 \\ \left(\frac{y-1}{3}\right)^{10} & 1 \leq y \leq 4 \\ 1 & y > 4 \end{cases}$$

(b) Find the probability that the maximum value in the sample is greater than 3.75

(1)

(c) (i) Sketch the probability density function of M for $1 \leq y \leq 4$

(ii) Write down the mode of M

(3)

(d) Use algebraic integration to find the exact value of $E(M)$

(4)



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Question 5 continued

Lined area for writing the answer to Question 5.



Question 5 continued

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Question 5 continued

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(Total for Question 5 is 11 marks)



6. Xiang owns a takeaway restaurant and wants to employ a delivery company. Xiang uses two companies, A and B , for one week each and records the delivery rate, x minutes per km, for each delivery.

Company A made 61 deliveries with $\bar{x}_A = 3.7$ and $s_A^2 = 1.64$

Company B made 51 deliveries with $\sum x_B = 142.8$ and $\sum x_B^2 = 461.34$

- (a) Find unbiased estimates for the mean and variance of the delivery rate for company B .

(3)

Xiang wants to carry out a two-sample t -test to determine whether there is a difference in the mean delivery rate for these companies.

- (b) State any assumptions Xiang must make about the delivery rates of the two companies.

(1)

- (c) Using a 5% level of significance, carry out Xiang's test. You should clearly state the hypotheses, test statistic and critical value you use.

(7)

Xiang noticed that a greater proportion of company A 's deliveries were in the town, with lots of traffic, whereas a greater proportion of company B 's deliveries were rural, with less traffic.

Xiang decides to carry out a further test. He selects a random sample of 10 customers who regularly place an order at the same time each week. He asks company A to deliver these orders one week and company B the next week. There were no roadworks or other incidents affecting travel time in either week.

The delivery rates are given in the table below.

Customer	1	2	3	4	5	6	7	8	9	10
Company A	5.4	3.6	2.5	4.7	2.3	3.2	3.4	4.1	5.9	4.5
Company B	4.9	3.9	2.3	4.5	2.6	2.8	3.1	3.8	5.2	4.3

- (d) Use a 5% level of significance to carry out a suitable test to determine whether there is a difference in the delivery rates of the two companies.

You should clearly state your hypotheses, test statistic and critical value.

(6)



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Question 6 continued

Lined area for writing the answer to Question 6.



Question 6 continued

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Question 6 continued

Lined area for writing answers.

(Total for Question 6 is 17 marks)



7. A random sample X_1, X_2, X_3, X_4 and X_5 is taken from a distribution $X \sim N(\mu, \sigma^2)$

Nina and Jonah each use unbiased estimators to estimate μ

Nina uses $\bar{X} = \frac{1}{5} \sum_{r=1}^5 X_r$ but Jonah uses $Y = \frac{X_1 + X_5}{2}$

(a) Find $P(\bar{X} - Y > \sigma)$

(7)

Nina and Jonah both calculated 95% confidence intervals for μ

They used their estimates and the normal distribution with the same value of σ

Their intervals were

(21.823, 38.177) and (27.029, 37.371)

(b) State, giving a reason, which of these intervals Nina calculated.

(1)

(c) Using one of these intervals, find the value of σ

(3)



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Question 7 continued

Lined area for writing the answer to Question 7.



Question 7 continued

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(Total for Question 7 is 11 marks)

TOTAL FOR PAPER IS 75 MARKS



Please check the examination details below before entering your candidate information

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Pearson Edexcel Level 3 GCE

Monday 24 June 2024

Afternoon (Time: 1 hour 30 minutes) **Paper reference** **9FM0/4B**

Further Mathematics
Advanced
PAPER 4B: Further Statistics 2

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 Mathematical Formulae and Statistical Tables (Green), calculator

Total Marks

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Pearson

1. Two students are experimenting with some water in a plastic bottle. The bottle is filled with water and a hole is put in the bottom of the bottle. The students record the time, t seconds, it takes for the water level to fall to each of 10 given values of the height, h cm, above the hole.

Student A models the data with an equation of the form $t = a + b\sqrt{h}$

The data is coded using $v = t - 40$ and $w = \sqrt{h}$ and the following information is obtained.

$$\sum v = 626 \quad \sum v^2 = 64678 \quad \sum w = 22.47 \quad S_{ww} = 4.52 \quad S_{vw} = -338.83$$

- (a) Find the equation of the regression line of t on $\sqrt{h}$ in the form $t = a + b\sqrt{h}$ (4)

The time it takes the water level to fall to a height of 9 cm above the hole is 47 seconds.

- (b) Calculate the residual for this data point.
Give your answer to 2 decimal places. (2)

Given that the residual sum of squares (RSS) for the model of t on $\sqrt{h}$ is the same as the RSS for the model of v on w ,

- (c) calculate the RSS for these 10 data points. (2)

Student B models the data with an equation of the form $t = c + dh$

The regression line of t on h is calculated and the residual sum of squares (RSS) is found to be 980 to 3 significant figures.

- (d) With reference to part (c) state, giving a reason, whether Student B 's model or Student A 's model is the more suitable for these data. (1)



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Question 1 continued

Lined area for writing the answer to Question 1.



Question 1 continued

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Question 1 continued

Lined area for writing answers.

(Total for Question 1 is 9 marks)



2. An estate agent asks customers to rank 7 features of a house, A, B, C, D, E, F and G , in order of importance. The responses for two randomly selected customers are in the table below.

Rank	1	2	3	4	5	6	7
Customer 1	A	E	C	F	G	B	D
Customer 2	E	F	C	G	A	D	B

- (a) Calculate Spearman's rank correlation coefficient for these data. (4)
- (b) Stating your hypotheses and critical value clearly, test at the 5% level of significance, whether or not the two customers are generally in agreement. (3)



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Question 2 continued

Lined area for writing answers.

(Total for Question 2 is 7 marks)



3. A factory produces bolts. The lengths of the bolts are normally distributed with mean μ mm and standard deviation 0.868 mm

A random sample of 15 of these bolts is taken and the mean length is 30.03 mm

- (a) Calculate a 90% confidence interval for μ

(3)

A suitable test, at the 10% level of significance, is carried out using these 15 bolts, to see whether or not there is evidence that the variance of the length of the bolts has increased.

- (b) Calculate the critical region for S^2

(3)

The manager of the factory decides that, in future, he will check each month whether the machine making the bolts is working properly. He uses a 10% level of significance to test whether or not there is evidence that

- the mean length of the bolts has changed
- the variance of the length of the bolts has increased

The next month a random sample of 15 bolts is taken.

The mean length of these bolts is 30.06 mm and the standard deviation is 1.02 mm

- (c) With reference to your answers to part (a) and part (b), state whether or not there is any evidence that the machine is **not** working properly.
Give reasons for your answer.

(2)



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Question 3 continued

Lined area for writing the answer to Question 3.

(Total for Question 3 is 8 marks)



- (1)

The random variables G and H are independent and $E(H) = 10$

- (3)

$$y = -4|x| + k$$

(5)

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Question 4 continued

Lined area for writing the answer to Question 4.



Question 4 continued

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Question 4 continued

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(Total for Question 4 is 9 marks)



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- 5.** A continuous random variable X has probability density function

$$f(x) = \begin{cases} ax^{-2} - bx^{-3} & 2 \leq x < \infty \\ 0 & \text{otherwise} \end{cases}$$

where a and b are constants.

Given that $P(X \leq 4) = \frac{3}{8}$

- (a) use algebraic integration to show that $a = 3$
Show your working clearly.

(6)

- (b) Find the exact value of the median of X

(4)



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Question 5 continued

Lined area for writing the answer to Question 5.



Question 5 continued

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Question 5 continued

Lined area for writing answers.

(Total for Question 5 is 10 marks)



6. A researcher set up a trial to assess the effect that a food supplement has on the increase in weight of Herdwick lambs. The researcher randomly selected 8 sets of twin lambs. One of each set of twins was given the food supplement and the other had no food supplement. The gain in weight, in kg, of each lamb over the period of the trial was recorded.

	Set of twin lambs	<i>A</i>	<i>B</i>	<i>C</i>	<i>D</i>	<i>E</i>	<i>F</i>	<i>G</i>	<i>H</i>
Weight gain (kg)	With food supplement	4.1	5.3	6.0	3.6	5.9	4.2	7.1	6.4
	No food supplement	5.0	4.8	5.2	3.4	5.1	3.9	7.0	6.5

- (a) State why a two sample t -test is not suitable for use with these data. (1)
- (b) Suggest 2 other factors about the lambs that the researcher may need to control when selecting the sample. (2)
- (c) State one assumption, in context, that needs to be made for a paired t -test to be valid. (1)

For a pair of twin lambs, the random variable W represents the weight gain of the lamb given the food supplement minus the weight gain of the lamb not given the food supplement.

- (d) Using the data in the table, calculate a 98% confidence interval for the mean of W . Show your working clearly. (5)

The researcher believes that the mean of W is greater than 200 g

- (e) Stating your hypotheses clearly, use your confidence interval to explain whether or not there is evidence to support the researcher's belief. (3)



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Question 6 continued

Lined area for writing the answer to Question 6.



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Question 6 continued

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Question 6 continued

Lined area for writing answers.

(Total for Question 6 is 12 marks)



7. Two organisations are each asked to carry out a survey to find out the proportion, p , of the population that would vote for a particular political party.

The first organisation finds that out of m people, X would vote for this particular political party.

The second organisation finds that out of n people, Y would vote for this particular political party.

An unbiased estimator, Q , of p is proposed where

$$Q = k \left(\frac{X}{m} + \frac{Y}{n} \right)$$

- (a) Show that $k = \frac{1}{2}$ (2)

A second unbiased estimator, R , of p is proposed where

$$R = \frac{aX}{m} + \frac{bY}{n}$$

- (b) Show that $a + b = 1$ (2)

Given that $m = 100$ and $n = 200$ and that R is a better estimator of p than Q

- (c) calculate the range of possible values of a
Show your working clearly.



DO NOT WRITE IN THIS AREA

Question 7 continued

Lined area for writing the answer to Question 7.



Question 7 continued

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Question 7 continued

Lined area for writing answers.

(Total for Question 7 is 11 marks)



8. A company packs chickpeas into small bags and large bags.

The weight of a small bag of chickpeas is normally distributed with mean 500 g and standard deviation 5 g

A random sample of 3 small bags of chickpeas is taken.

- (a) Find the probability that the total weight of these 3 bags of chickpeas is between 1490 g and 1530 g

(3)

The weight of a large bag of chickpeas is normally distributed with mean 1020 g and standard deviation 20 g

One large bag and one small bag of chickpeas are chosen at random.

- (b) Calculate the probability that the weight of the large bag of chickpeas is at least 30 g more than twice the weight of the small bag of chickpeas.
Show your working clearly.

(6)



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Question 8 continued

Lined area for writing the answer to Question 8.



Question 8 continued

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(Total for Question 8 is 9 marks)

TOTAL FOR PAPER IS 75 MARKS



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Pearson Edexcel Level 3 GCE

Monday 26 June 2023

Afternoon (Time: 1 hour 30 Minutes) **Paper reference** **9FM0/4B**

Further Mathematics

Advanced

PAPER 4B: Further Statistics 2

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Total Marks

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Turn over ►

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Pearson

1. Baako is investigating the times taken by children to run a 100 m race, x seconds, and a 500 m race, y seconds. For a sample of 20 children, Baako obtains the time taken by each child to run each race.

Here are Baako's summary statistics.

$$S_{xx} = 314.55 \quad S_{yy} = 9026 \quad S_{xy} = 1610$$

$$\bar{x} = 19.65 \quad \bar{y} = 108$$

- (a) Calculate the product moment correlation coefficient between the times taken to run the 100 m race and the times taken to run the 500 m race.

(2)

- (b) Show that the equation of the regression line of y on x can be written as

$$y = 5.12x + 7.42$$

where the gradient and y intercept are given to 3 significant figures.

(3)

The child who completed the 100 m race in 20 seconds took 104 seconds to complete the 500 m race.

- (c) Find the residual for this child.

(1)

The table below shows the signs of the residuals for the 20 children in order of finishing time for the 100 m race.

Sign of residual	+	+	+	+	-	-	+	-	-	-	-	-	-	-	+	+	+	+	+
-------------------------	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---

- (d) Explain what the signs of the residuals show about the model's predictions of the 500 m race times for the children who are fastest and slowest over the 100 m race.

(1)



DO NOT WRITE IN THIS AREA

Question 1 continued

Lined area for writing the answer to Question 1.

(Total for Question 1 is 7 marks)



2. Camilo grows two types of apple, green apples and red apples.

The standard deviation of the weights of green apples is known to be 3.5 grams.

A random sample of 80 green apples has a mean weight of 128 grams.

- (a) Find a 98% confidence interval for the mean weight of the population of green apples.
Show your working clearly and give the confidence interval limits to 2 decimal places.

(3)

Camilo believes that the mean weight of the population of green apples is more than 10 grams greater than the mean weight of the population of red apples.

A random sample of n red apples has a mean weight of 117 grams.

The standard deviation of the weights of the red apples is known to be 4 grams.

A test of Camilo's belief is carried out at the 5% level of significance.

- (b) State the null and alternative hypotheses for this test.

(1)

- (c) Find the smallest value of n for which the null hypothesis will be rejected.

(6)

- (d) Explain the relevance of the Central Limit Theorem in parts (a) and (c).

(1)

- (e) Given that $n = 85$, state the conclusion of the hypothesis test.

(1)



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Question 2 continued

Lined area for writing the answer to Question 2.



Question 2 continued

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DO NOT WRITE IN THIS AREA

Question 2 continued

Lined area for writing answers.

(Total for Question 2 is 12 marks)



3. Two machines, A and B , are used to fill bottles of water. The amount of water dispensed by each machine is normally distributed.

Samples are taken from each machine and the amount of water, x ml, dispensed in each bottle is recorded. The table shows the summary statistics for Machine A .

	Sample size	$\sum x$	$\sum x^2$
Machine A	9	2268	571 700

- (a) Find a 95% confidence interval for the variance of the amount of water dispensed in each bottle by Machine A .

(4)

For Machine B , a random sample of 11 bottles is taken. The sample variance of the amount of water dispensed in bottles is 12.7 ml^2

- (b) Test, at the 10% level of significance, whether there is evidence that the variances of the amounts of water dispensed in bottles by the two machines are different.

You should state the hypotheses and the critical value used.

(4)



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Question 3 continued

Lined area for writing the answer to Question 3.

(Total for Question 3 is 8 marks)

- Large boxes of eggs contain 12 randomly selected eggs.

- (3)

- (5)

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Question 4 continued

Lined area for writing answers.

(Total for Question 4 is 8 marks)



- The table shows the percentage of words recalled by each person before and after the training.

Person	<i>A</i>	<i>B</i>	<i>C</i>	<i>D</i>	<i>E</i>	<i>F</i>	<i>G</i>	<i>H</i>
Percentage of words recalled before training	24	33	33	39	30	38	32	34
Percentage of words recalled after training	28	30	37	41	32	44	35	34

- (a) State why a paired t -test is suitable for these data. (1)
- (b) State an assumption that needs to be made in order to carry out a paired t -test in this case. (1)
- (c) Test, at the 5% level of significance, whether or not there is evidence of an increase in the percentage of words recalled after receiving the psychologist's training. State your hypotheses, test statistic and critical value used for this test. (7)

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Question 5 continued

Lined area for writing answers.

(Total for Question 5 is 9 marks)



DO NOT WRITE IN THIS AREA

Question 6 continued

Lined area for writing the answer to Question 6.

(Total for Question 6 is 10 marks)



DO NOT WRITE IN THIS AREA

Question 7 continued

Handwriting practice area with horizontal lines.

(Total for Question 7 is 9 marks)



8. A bag contains a large number of marbles of which an unknown proportion, p , is yellow.

Three random samples of size n are taken, and the number of yellow marbles in each sample, Y_1 , Y_2 and Y_3 , is recorded.

Two estimators $\hat{p}_1$ and $\hat{p}_2$ are proposed to estimate the value of p

$$\hat{p}_1 = \frac{Y_1 + 3Y_2 - 2Y_3}{2n}$$

$$\hat{p}_2 = \frac{2Y_1 + 3Y_2 + Y_3}{6n}$$

- (a) Show that $\hat{p}_1$ and $\hat{p}_2$ are both unbiased estimators of p (3)

- (b) Find the variance of $\hat{p}_1$

The variance of $\hat{p}_2$ is $\frac{7p(1-p)}{18n}$

- (c) State, giving a reason, which is the better estimator. (2)

The estimator $\hat{p}_3 = \frac{Y_1 + aY_2 + 3Y_3}{bn}$ where a and b are positive integers.

- (d) Find the pair of values of a and b such that $\hat{p}_3$ is a better unbiased estimator of p than both $\hat{p}_1$ and $\hat{p}_2$.
You must show all stages of your working.
- (5)**



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Question 8 continued

Lined area for writing the answer to Question 8.



Question 8 continued

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(Total for Question 8 is 12 marks)

TOTAL FOR PAPER IS 75 MARKS



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Pearson Edexcel Level 3 GCE

Time 1 hour 30 minutes

Paper
reference

9FM0/4B

Further Mathematics

Advanced

PAPER 4B: Further Statistics 2

You must have:

Mathematical Formulae and Statistical Tables (Green), calculator

Total Marks

Candidates may use any calculator permitted by Pearson regulations. Calculators must not have the facility for symbolic algebra manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions

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- You should show sufficient working to make your methods clear.
Answers without working may not gain full credit.
- Values from statistical tables should be quoted in full. If a calculator is used instead of the tables the value should be given to an equivalent degree of accuracy.
- Inexact answers should be given to three significant figures unless otherwise stated.

Information

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- There are 8 questions in this question paper. The total mark for this paper is 75.
- The marks for **each** question are shown in brackets
– *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ►

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1. Kwame is investigating a possible relationship between average March temperature, $t^{\circ}\text{C}$, and tea yield, y kg/hectare, for tea grown in a particular location. He uses 30 years of past data to produce the following summary statistics for a linear regression model, with tea yield as the dependent variable.

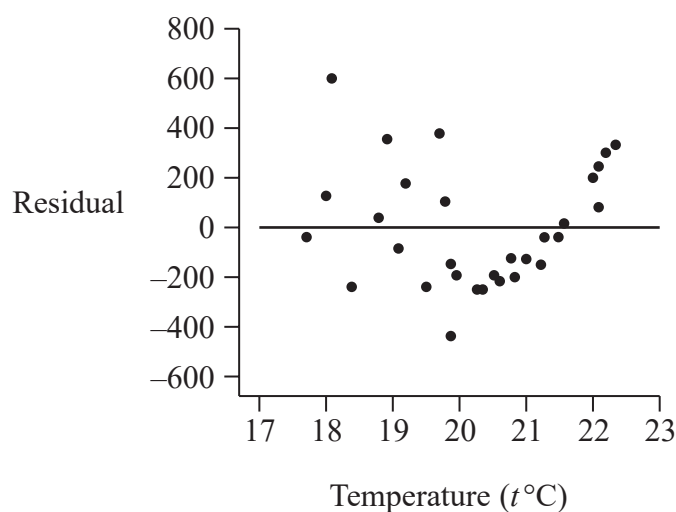
$$\text{Residual Sum of Squares (RSS)} = 1\,666\,567 \quad S_{tt} = 52.0 \quad S_{yy} = 1\,774\,155$$

$$\text{least squares regression line:} \quad \text{gradient} = 45.5 \quad y\text{-intercept} = 2080$$

- (a) Use the regression model to predict the tea yield for an average March temperature of 20°C

(1)

He also produces the following residual plot for the data.



- (b) Explain what you understand by the term residual.

(1)

- (c) Calculate the product moment correlation coefficient between t and y

(2)

- (d) Explain why the linear model may not be a good fit for the data

(i) with reference to your answer to part (c)

(ii) with reference to the residual plot.

(2)

Question 1 continues on page 4



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Question 1 continued

Lined area for writing the answer to Question 1.



Question 1 continued

Kwame also collects data on total March rainfall, w mm, for each of these 30 years.

For a linear regression model of w on t the following summary statistic is found.

$$\text{Residual Sum of Squares (RSS)} = 86\,754$$

Kwame concludes that since this model has a smaller RSS, there must be a stronger linear relationship between w and t than between y and t (where $\text{RSS} = 1\,666\,567$)

- (e) State, giving a reason, whether or not you agree with the reasoning that led to Kwame's conclusion.

(1)



DO NOT WRITE IN THIS AREA

Question 1 continued

Handwriting practice area with horizontal lines.

(Total for Question 1 is 7 marks)



2. A factory produces yellow tennis balls and white tennis balls. Independent samples, one of yellow tennis balls and one of white tennis balls, are taken. The table shows information about the weights of the yellow tennis balls, Y grams, and the weights of the white tennis balls, W grams.

	Sample size	Mean weight of random sample (grams)	Known population standard deviation of weights (grams)
Yellow tennis balls	120	57.2	1.2
White tennis balls	140	56.9	0.9

- (a) Find a 95% confidence interval for the mean weight of **yellow** tennis balls. (3)

Jamie claims that the mean weight of the population of yellow tennis balls is greater than the mean weight of the population of white tennis balls. A test of Jamie's claim is carried out.

- (b) (i) Specify the approximate distribution of $\bar{Y} - \bar{W}$ under the null hypothesis of the test. (3)

- (ii) Explain the relevance of the large sample sizes to your answer to part (i). (1)

- (c) Complete the hypothesis test using a 5% level of significance.
You should state your hypotheses and the value of your test statistic clearly.

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Question 2 continued

Lined area for writing the answer to Question 2.



Question 2 continued

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Question 2 continued

Lined area for writing answers.

(Total for Question 2 is 12 marks)



3. The random variable $X \sim N(5, 0.4^2)$ and the random variable $Y \sim N(8, 0.1^2)$

X and Y are independent random variables.

A random sample of a independent observations is taken from the distribution of X and one observation is taken from the distribution of Y

The random variable $W = X_1 + X_2 + X_3 + \dots + X_a + bY$ and has the distribution $N(169, 2^2)$

Find the value of a and the value of b

(6)



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Question 3 continued

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(Total for Question 3 is 6 marks)



4. A doctor believes that a four-week exercise programme can reduce the resting heart rate of her patients. She takes a random sample of 7 patients and records their resting heart rate before the exercise programme and again after the exercise programme.

Patient	<i>A</i>	<i>B</i>	<i>C</i>	<i>D</i>	<i>E</i>	<i>F</i>	<i>G</i>
Resting heart rate before	65	68	77	79	80	88	92
Resting heart rate after	63	65	73	76	80	84	80

- (a) Using a 5% level of significance, carry out an appropriate test of the doctor's belief. You should state your hypotheses, test statistic and critical value.

(7)

- (b) State the assumption made about the resting heart rates that was required to carry out the test.

(1)



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Question 4 continued

Lined area for writing answers.

(Total for Question 4 is 8 marks)



- 5.** The concentration of an air pollutant is measured in micrograms/m³

Samples of air were taken at two different sites and the concentration of this particular air pollutant was recorded.

For Site *A* the summary statistics are shown below.

	number of samples	s_A^2
Site A	13	6.39

For Site *B* there were 9 samples of air taken.

A test of the hypothesis $H_0: \sigma_A^2 = \sigma_B^2$ against the hypothesis $H_1: \sigma_A^2 \neq \sigma_B^2$ is carried out using a 2% level of significance.

- (a) State a necessary assumption required to carry out the test. (1)

Given that the assumption in part (a) holds,

- (b) find the set of values of s_B^2 that would lead to the null hypothesis being rejected, (4)
- (c) find a 99% confidence interval for the variance of the concentration of the air pollutant at **Site A**. (3)



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Question 5 continued

Lined area for writing answers.

(Total for Question 5 is 8 marks)



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Question 6 continued

Lined area for writing the answer to Question 6.



Question 6 continued

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Question 6 continued

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(Total for Question 6 is 15 marks)



7. A rectangle is to have an area of 40 cm^2

The length of the rectangle, $L \text{ cm}$, follows a continuous uniform distribution over the interval $[4, 10]$

Find the expected value of the perimeter of the rectangle.

Use algebraic integration, rather than your calculator, to evaluate any definite integrals.

(7)



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Question 7 continued

Handwriting practice area with horizontal lines.

(Total for Question 7 is 7 marks)



8. The continuous random variable X has cumulative distribution function given by

$$F(x) = \begin{cases} 0 & x < 1 \\ 1.5x - 0.25x^2 - 1.25 & 1 \leq x \leq 3 \\ 1 & x > 3 \end{cases}$$

- (a) Find the exact value of the median of X (2)

- (b) Find $P(X < 1.6 | X > 1.2)$ (3)

The random variable $Y = \frac{1}{X}$

- (c) Specify fully the cumulative distribution function of Y (4)

- (d) Hence or otherwise find the mode of Y (3)



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Question 8 continued

Lined area for writing the answer to Question 8.



Question 8 continued

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(Total for Question 8 is 12 marks)

TOTAL FOR PAPER IS 75 MARKS



Please check the examination details below before entering your candidate information

Candidate surname		Other names	
Pearson Edexcel		Centre Number	Candidate Number
Level 3 GCE			
Time 1 hour 30 minutes	Paper reference	9FM0/4B	
Further Mathematics Advanced PAPER 4B: Further Statistics 2			
You must have: Mathematical Formulae and Statistical Tables (Green), calculator			Total Marks

Candidates may use any calculator permitted by Pearson regulations. Calculators must not have the facility for algebraic manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

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Information

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- There are 7 questions in this question paper. The total mark for this paper is 75.
- The marks for **each** question are shown in brackets
– *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.
- Good luck with your examination.

Turn over ►

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1. Anisa is investigating the relationship between marks on a History test and marks on a Geography test. She collects information from 7 students. She wants to calculate the Spearman's rank correlation coefficient for the 7 students so she ranks their performance on each test.

Student	History mark	Geography mark	History rank	Geography rank
<i>A</i>	76	58	1	3
<i>B</i>	70	60	2	2
<i>C</i>	64	57	<i>s</i>	<i>t</i>
<i>D</i>	64	63	<i>s</i>	1
<i>E</i>	64	57	<i>s</i>	<i>t</i>
<i>F</i>	59	50	6	7
<i>G</i>	55	52	7	6

- (a) Write down the value of *s* and the value of *t*

(2)

The full product moment correlation coefficient (pmcc) formula is used with the ranks

to calculate the Spearman's rank correlation coefficient instead of $r_s = 1 - \frac{6\sum d^2}{n(n^2 - 1)}$

and the value obtained is 0.7106 to 4 significant figures.

- (b) Explain why the full pmcc formula is used to carry out the calculation.

(1)

- (c) Stating your hypotheses clearly, test whether or not there is evidence to suggest that the higher a student ranks in the History test, the higher the student ranks in the Geography test. Use a 5% level of significance.

(4)



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Question 1 continued

Lined area for writing answers.

(Total for Question 1 is 7 marks)



2. A company produces two colours of candles, blue and white. The standard deviation of the burning times of the blue candles is 2.6 minutes and the standard deviation of the burning times of the white candles is 2.4 minutes.

Nissim claims that the mean burning time of blue candles is more than 5 minutes greater than the mean burning time of white candles.

A random sample of 90 blue candles is found to have a mean burning time of 39.5 minutes. A random sample of 80 white candles is found to have a mean burning time of 33.7 minutes.

- (a) Stating your hypotheses clearly, use a suitable test to assess Nissim's belief. Use a 1% level of significance.

(6)

- (b) Explain how the hypothesis test in part (a) would be carried out differently if the variances of the burning times of candles were unknown.

(1)

The burning times for the candles may not follow a normal distribution.

- (c) Describe the effect this would have on the calculations in the hypothesis test in part (a). Give a reason for your answer.

(2)



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Question 2 continued

Lined area for writing the answer to Question 2.



Question 2 continued

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Question 2 continued

Lined area for writing answers.

(Total for Question 2 is 9 marks)



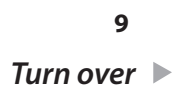
3. The continuous random variable X has cumulative distribution function given by

$$F(x) = \begin{cases} 0 & x < 2 \\ 1.25 - \frac{2.5}{x} & 2 \leq x \leq 10 \\ 1 & x > 10 \end{cases}$$

- (a) Find $P(\{X < 5\} \cup \{X > 8\})$ (2)
- (b) Find the median of X . (2)
- (c) Find $E(X^2)$ (3)
- (d) (i) Sketch the probability density function of X .
(ii) Describe the skewness of the distribution of X . (3)



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Question 3 continued

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Question 3 continued

Lined area for writing answers.

(Total for Question 3 is 10 marks)



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Question 4 continued

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Question 4 continued

Lined area for writing answers.

(Total for Question 4 is 10 marks)



5. The continuous random variable X is uniformly distributed over the interval $[0, 4\beta]$, where β is an unknown constant.

Three independent observations, X_1 , X_2 and X_3 , are taken of X and the following estimators for β are proposed

$$A = \frac{X_1 + X_2}{2}$$

$$B = \frac{X_1 + 2X_2 + 3X_3}{8}$$

$$C = \frac{X_1 + 2X_2 - X_3}{8}$$

- (a) Calculate the bias of A , the bias of B and the bias of C (5)
- (b) By calculating the variances, explain which of B or C is the better estimator for β (4)
- (c) Find an unbiased estimator for β (1)



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Question 5 continued

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Question 5 continued

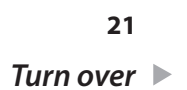
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(Total for Question 5 is 10 marks)



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Question 6 continued

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Question 6 continued

Lined area for writing answers.

(Total for Question 6 is 15 marks)



7. The weights of a particular type of apple, A grams, and a particular type of orange, R grams, each follow independent normal distributions.

$$A \sim N(160, 12^2) \quad R \sim N(140, 10^2)$$

(a) Find the distribution of

(i) $A + R$

(ii) the total weight of 2 randomly selected apples.

(3)

A box contains 4 apples and 1 orange only. Jesse selects 2 pieces of fruit at random from the box.

(b) Find the probability that the total weight of the 2 pieces of fruit exceeds 310 grams.

(3)

From a large number of apples and oranges, Celeste selects m apples and 1 orange at random. The random variable W is given by

$$W = \left(\sum_{i=1}^m A_i \right) - n \times R$$

where n is a positive integer.

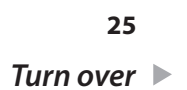
Given that the middle 95% of the distribution of W lies between 1100.08 and 1499.92 grams,

(c) find the value of m and the value of n

(8)



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Question 7 continued

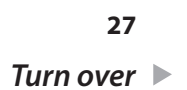
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Question 7 continued

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(Total for Question 7 is 14 marks)

TOTAL FOR PAPER IS 75 MARKS



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Level 3 GCE**

Centre Number

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Candidate Number

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Friday 23 October 2020

Afternoon (Time: 1 hour 30 minutes)

Paper Reference **9FM0/4B**

Further Mathematics

Advanced

Paper 4B: Further Statistics 2

You must have:

Mathematical Formulae and Statistical Tables (Green), calculator

Total Marks

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- Check your answers if you have time at the end.

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- 1 Gina receives a large number of packages from two companies, A and B . She believes that the variance of the weights of packages from company A is greater than the variance of the weights of packages from company B .

Gina takes a random sample of 7 packages from company A and an independent random sample of 10 packages from company B . Her results are summarised below

$$\bar{a} = 300 \quad S_{aa} = 145\,496 \quad \bar{b} = 233.4 \quad S_{bb} = 56\,364.4$$

[You may assume that the weights of packages from the two companies are normally distributed.]

Test Gina's belief. Use a 5% level of significance and state your hypotheses clearly.

(6)



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Question 1 continued

Lined area for writing answers.

(Total for Question 1 is 6 marks)



- 2 Jemima makes jam to sell in a local shop. The jam is sold in jars and the weight of jam in a jar is normally distributed.

Jemima takes a random sample of 8 of her jars of jam and weighs the contents of each jar, x grams. Her results are summarised as follows

$$\sum x = 3552 \quad \sum x^2 = 1577314$$

- (a) Calculate a 95% confidence interval for the mean weight of jam in a jar.

(5)

The labels on the jars state that the average contents weigh 440 grams.

- (b) State, giving a reason, whether or not Jemima should be concerned about the labels on her jars of jam.

(1)



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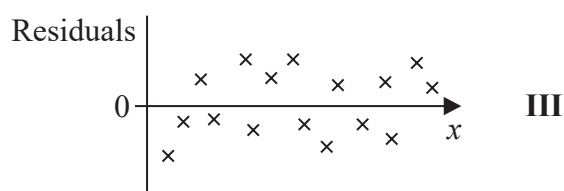
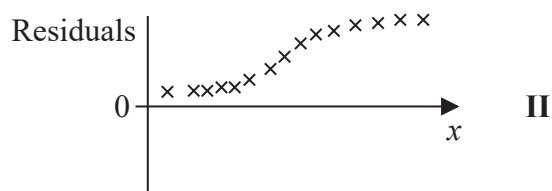
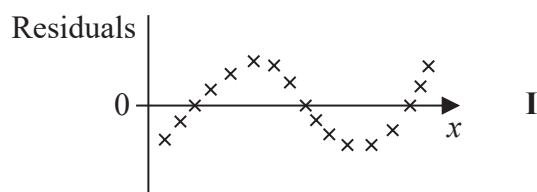
Question 2 continued

Lined area for writing the answer to Question 2.

(Total for Question 2 is 6 marks)



- 3 Below are 3 sketches from some students of the residuals from their linear regressions of y on x .



For each sketch you should state, giving your reason,

- (i) whether or not the sketch is feasible

and if it is feasible

- (ii) whether or not the sketch suggests a linear or a non-linear relationship between y and x .
(6)



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Question 3 continued

Lined area for writing answers.

(Total for Question 3 is 6 marks)



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Question 4 continued

Lined area for writing the answer to Question 4.



Question 4 continued

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Question 4 continued

Lined area for writing the answer to Question 4.

(Total for Question 4 is 7 marks)



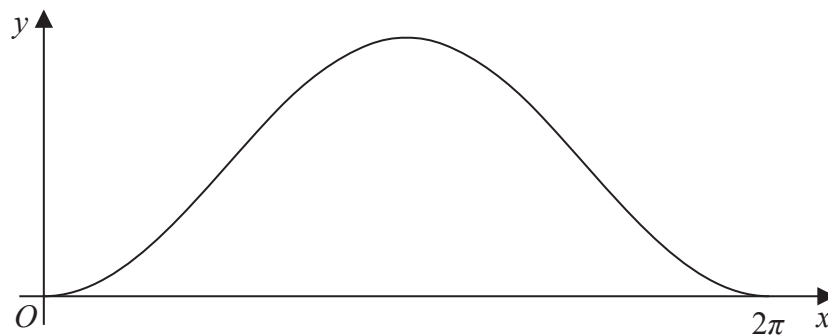


Figure 1

The random variable X has probability density function $f(x)$ and Figure 1 shows a sketch of $f(x)$ where

$$f(x) = \begin{cases} k(1 - \cos x) & 0 \leq x \leq 2\pi \\ 0 & \text{otherwise} \end{cases}$$

(a) Show that $k = \frac{1}{2\pi}$

(3)

The random variable $Y \sim N(\mu, \sigma^2)$ and $E(Y) = E(X)$

The probability density function of Y is $g(y)$, where

$$g(y) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{y-\mu}{\sigma}\right)^2} \quad -\infty < y < \infty$$

Given that $g(\mu) = f(\mu)$

(b) find the exact value of σ

(3)

(c) Calculate the error in using $P\left(\frac{\pi}{2} < Y < \frac{3\pi}{2}\right)$ as an approximation to $P\left(\frac{\pi}{2} < X < \frac{3\pi}{2}\right)$

(4)



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Question 5 continued

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Question 5 continued

Lined area for writing the answer to Question 5.

(Total for Question 5 is 10 marks)



- 6 A new employee, Kim, joins an existing employee, Jiang, to work in the quality control department of a company producing steel rods. Each day a random sample of rods is taken, their lengths measured and a 95% confidence interval for the mean length of the rods, in metres, is calculated. It is assumed that the lengths of the rods produced are normally distributed.

Kim took a random sample of 25 rods and used the t distribution to obtain a 95% confidence interval of (1.193, 1.367) for the mean length of the rods.

Jiang commented that this interval was a little wider than usual and explained that they usually assume that the standard deviation does not change and can be taken as 0.175 metres.

- (a) Test, at the 10% level of significance, whether or not Kim's sample suggests that the standard deviation is different from 0.175 metres. State your hypotheses clearly. (9)

Using Kim's sample and the normal distribution with a standard deviation of 0.175 metres,

- (b) find a 95% confidence interval for the mean length of the rods. (3)



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- 7 Fence panels come in two sizes, large and small. The lengths of the large panels are normally distributed with mean 198 cm and standard deviation 5 cm. The lengths of the small panels are normally distributed with mean 74 cm and standard deviation 3 cm.

- (a) Find the probability that the total length of a random sample of 3 large panels is greater than the total length of a random sample of 8 small panels. (6)

One large panel and one small panel are selected at random.

- (b) Find the probability that the length of the large panel is more than $\frac{8}{3}$ times the length of the small panel. (5)

Rosa needs 1000 cm of fencing. The large panels cost £80 each and the small panels cost £30 each. Rosa's plan is to buy 5 large panels and measure the total length. If the total length is less than 1000 cm she will then buy one small panel as well.

- (c) Calculate whether or not the expected cost of Rosa's plan is cheaper than simply buying 14 small panels. (6)



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Question 7 continued

Lined area for writing the answer to Question 7.



Question 7 continued

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Question 7 continued

Lined area for writing answers.

(Total for Question 7 is 17 marks)



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Question 8 continued

Lined area for writing the answer to Question 8.



Question 8 continued

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(Total for Question 8 is 11 marks)

TOTAL FOR PAPER IS 75 MARKS



Please check the examination details below before entering your candidate information			
Candidate surname		Other names	
Pearson Edexcel Level 3 GCE		Centre Number	Candidate Number
Tuesday 25 June 2019			
Morning (Time: 1 hour 30 minutes)		Paper Reference 9FM0/4B	
Further Mathematics Advanced Paper 4B: Further Statistics 2			
You must have: Mathematical Formulae and Statistical Tables (Green), calculator			Total Marks

Candidates may use any calculator permitted by Pearson regulations. Calculators must not have the facility for algebraic manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions

- Use **black** ink or ball-point pen.
- If pencil is used for diagrams/sketches/graphs it must be dark (HB or B).
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- Answer **all** questions and ensure that your answers to parts of questions are clearly labelled.
- Answer the questions in the spaces provided
– *there may be more space than you need.*
- You should show sufficient working to make your methods clear. Answers without working may not gain full credit.
- Answers should be given to three significant figures unless otherwise stated.

Information

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- There are 8 questions in this question paper. The total mark for this paper is 75.
- The marks for **each** question are shown in brackets
– *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ►

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Question 1 continued

Lined area for writing the answer to Question 1.

(Total for Question 1 is 8 marks)

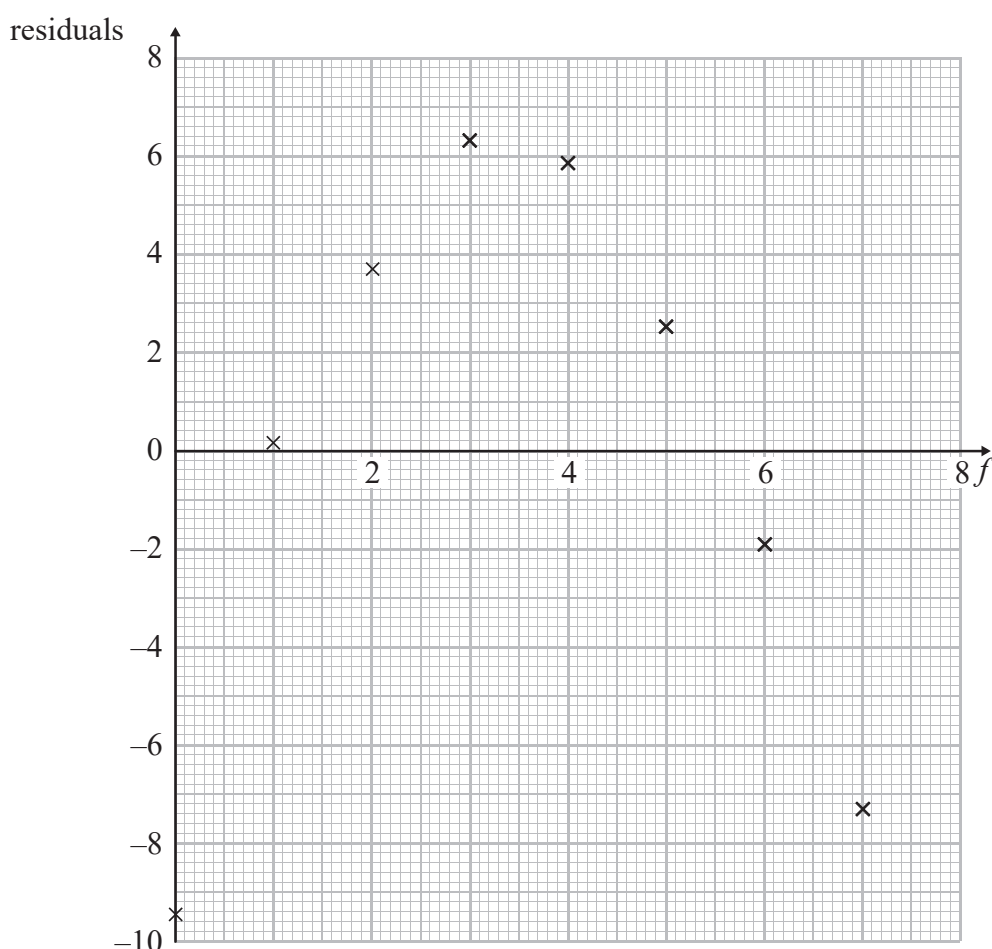


- 2 A large field of wheat is split into 8 plots of equal area. Each plot is treated with a different amount of fertiliser, f grams/m². The yield of wheat, w tonnes, from each plot is recorded. The results are summarised below.

$$\sum f = 28 \quad \sum w = 303 \quad \sum w^2 = 13\,447 \quad S_{ff} = 42 \quad S_{fw} = 269.5$$

- (a) Calculate the product moment correlation coefficient between f and w (2)
- (b) Interpret the value of your product moment correlation coefficient. (1)
- (c) Find the equation of the regression line of w on f in the form $w = a + bf$ (3)
- (d) Using your equation, estimate the decrease in yield when the amount of fertiliser decreases by 0.5 grams/m² (1)

The residuals of the data recorded are calculated and plotted on the graph below.



- (e) With reference to this graph, comment on the suitability of the model you found in part (c). (2)
- (f) Suggest how you might be able to refine your model. (1)



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Question 2 continued

Lined area for writing the answer to Question 2.



Question 2 continued

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Question 2 continued

Lined area for writing answers.

(Total for Question 2 is 10 marks)



- 3 Yin grows two varieties of potato, plant A and plant B . A random sample of each variety of potato is taken and the yield, x kg, produced by each plant is measured. The following statistics are obtained from the data.

	Number of plants	$\sum x$	$\sum x^2$
A	25	194.7	1637.37
B	26	227.5	2031.19

- (a) Stating your hypotheses clearly, test, at the 10% significance level, whether or not the variances of the yields of the two varieties of potato are the same.

(7)

- (b) State an assumption you have made in order to carry out the test in part (a).

(1)



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Question 3 continued

Lined area for writing the answer to Question 3.

(Total for Question 3 is 8 marks)



- $$F(x) = \begin{cases} 0 & x \leq 0 \\ k\left(x^3 - \frac{3}{8}x^4\right) & 0 < x \leq 2 \\ 1 & x > 2 \end{cases}$$

(a) Show that $k = \frac{1}{2}$ (1)

- $$(ii) \text{ the mode of } X \tag{6}$$

- (c) Describe, giving a reason, the skewness of the distribution of X (1)

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Question 4 continued

Lined area for writing the answer to Question 4.



Question 4 continued

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Question 4 continued

Lined area for writing the answer to Question 4.

(Total for Question 4 is 8 marks)



- 5 Alexa believes that students are equally likely to achieve the same percentage score on each of two tests, paper I and paper II. She randomly selects 8 students and gives them each paper I and paper II. The percentage scores for each paper are recorded.

The following paired data are collected.

Student	<i>A</i>	<i>B</i>	<i>C</i>	<i>D</i>	<i>E</i>	<i>F</i>	<i>G</i>	<i>H</i>
Paper I (%)	70	70	84	80	64	65	65	90
Paper II (%)	64	76	72	74	68	64	58	76

Test, at the 1% significance level, whether or not there is evidence to support Alexa's belief. State your hypotheses clearly and show your working.

(7)



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Question 5 continued

Lined area for writing the answer to Question 5.

(Total for Question 5 is 7 marks)



- 6 A company manufactures bolts. The diameter of the bolts follows a normal distribution with a mean diameter of 5 mm.

Stan believes that the mean diameter of the bolts is less than 5 mm. He takes a random sample of 10 bolts and measures their diameters. He calculates some statistics but spills ink on his work before completing them. The only information he has left is as follows

4.5 4.5 5.5 4.8 4.9 4.7 5
 $X \sim N(5$
 $\sum x = 48.4$
 $\bar{x} =$
99% confidence interval for the variance is = (0.01712, 0.23280)

Stating your hypotheses clearly, test, at the 5% level of significance, whether or not Stan's belief is supported.

(9)



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Question 6 continued

Lined area for writing answers.

(Total for Question 6 is 9 marks)



- 7 A manufacturer makes two versions of a toy. One version is made out of wood and the other is made out of plastic.

The weights, W kg, of the wooden toys are normally distributed with mean 2.5 kg and standard deviation 0.7 kg. The weights, X kg, of the plastic toys are normally distributed with mean 1.27 kg and standard deviation 0.4 kg. The random variables W and X are independent.

- (a) Find the probability that the weight of a randomly chosen wooden toy is more than double the weight of a randomly chosen plastic toy.

(6)

The manufacturer packs n of these wooden toys and $2n$ of these plastic toys into the same container. The maximum weight the container can hold is 252 kg.

The probability of the contents of this container being overweight is 0.2119 to 4 decimal places.

- (b) Calculate the value of n .

(8)



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Question 7 continued

Lined area for writing the answer to Question 7.



Question 7 continued

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Question 7 continued

Lined area for writing answers.

(Total for Question 7 is 14 marks)



- 8 Nine athletes, A, B, C, D, E, F, G, H and I , competed in both the 100m sprint and the long jump. After the two events the positions of each athlete were recorded and Spearman's rank correlation coefficient was calculated and found to be 0.85

- (a) Stating your hypotheses clearly, test whether or not there is evidence to suggest that the higher an athlete's position is in the 100m sprint, the higher their position is in the long jump. Use a 5% level of significance.

(4)

The piece of paper the positions were recorded on was mislaid. Although some of the athletes agreed their positions, there was some disagreement between athletes B, C and D over their long jump results.

The table shows the results that are agreed to be correct.

Athlete	A	B	C	D	E	F	G	H	I
Position in 100m sprint	4	6	7	9	2	8	3	1	5
Position in long jump	5				4	9	3	1	2

Given that there were no tied ranks,

- (b) find the correct positions of athletes B, C and D in the long jump. You must show your working clearly and give reasons for your answers.

(5)

- (c) Without recalculating the coefficient, explain how Spearman's rank correlation coefficient would change if athlete H was disqualified from both the 100m sprint and the long jump.

(2)



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Question 8 continued

Lined area for writing the answer to Question 8.



Question 8 continued

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(Total for Question 8 is 11 marks)

TOTAL FOR PAPER IS 75 MARKS



Please check the examination details below before entering your candidate information

Candidate surname		Other names	
Pearson Edexcel Level 3 GCE		Centre Number	Candidate Number
Mock Paper Set 1			
(Time: 1 hour 30 minutes)		Paper Reference 9FM0/4B	
Further Mathematics Advanced Paper 4B: Further Statistics 2			
You must have: Mathematical Formulae and Statistical Tables, calculator			Total Marks

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Advice

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Answer ALL questions. Write your answers in the spaces provided.

1. A company imports consumer goods from around the world. When the company receives a new shipment they need staff to work overtime.

The finance department wishes to investigate the relationship between the size of the shipment, x containers, and the wage bill, y in hundreds of pounds.

Data were collected for a random sample of 10 weeks when new shipments were received and are summarised by the following statistics.

$$\sum xy = 604430 \quad \bar{x} = 326 \quad \bar{y} = 180.8 \quad \sigma_x = 139.2 \quad \sigma_y = 11.1$$

- (a) Calculate the product moment correlation coefficient for these data. (4)
- (b) Stating your hypotheses clearly, test whether there is evidence of correlation between the size of the shipment and wage bill. Use a 5% level of significance. (4)



DO NOT WRITE IN THIS AREA

Question 1 continued

Lined area for writing answers.

(Total for Question 1 is 8 marks)



2. A finance manager records the number of lorry loads of goods, x , leaving his factory each week, together with the total production costs, c , measured in £100s, for those goods.

He takes a random sample of 12 weeks and his results are summarised below.

$$\sum c = 783 \quad \sum x = 540 \quad S_{xc} = 6452 \quad \sum x^2 = 30\,786 \quad \sum c^2 = 57\,557$$

- (a) Find the equation of the regression line of c on x in the form $c = a + bx$ as a model for these results. (4)
- (b) According to this model, give an interpretation of
- (i) the value of b
 - (ii) the value of a
- (2)

In the finance manager's sample, the actual total production cost for a week with 72 lorry loads was £8900 and for a week with 12 lorry loads the total production cost was £3500

- (c) Calculate the residuals for these 2 weeks. (3)
- (d) Calculate the residual sum of squares for these 12 weeks. (3)
- (e) Using your answer to part (d), suggest an approximate range of values for the total production costs in a week when 60 lorry loads leave the factory. (2)

The managing director believes that if the factory produces more goods, then the production cost per lorry load should decrease.

- (f) Comment on whether or not the managing director's belief is consistent with the finance manager's model. (1)



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Question 2 continued

Lined area for writing the answer to Question 2.



Question 2 continued

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Question 2 continued

Lined area for writing answers.

(Total for Question 2 is 15 marks)



3. A company manufactures ballpoint pens. The pens come in two colours, black and red. The company uses a machine that assesses the life of a pen by measuring the length of a continuous line drawn using the pen.

An experiment was conducted to compare the life of black and red pens. In the experiment a random sample of 80 black pens and a random sample of 75 red pens were tested.

The table below summarises the results of the test.

Pen Colour	Black	Red
Mean line length (km)	1.21	1.25
Standard deviation (km)	0.15	0.13

You may assume that the length of a continuous line drawn by a randomly selected pen is normally distributed.

- (a) Find a 90% confidence interval for the mean line length of black pens. (3)
- (b) Using a 5% significance level, test whether the mean line length of black pens differs significantly from the mean line length of red pens. State your hypotheses clearly. (6)



DO NOT WRITE IN THIS AREA

Question 3 continued

Lined area for writing answers.

(Total for Question 3 is 9 marks)



4. In a newspaper article a journalist claimed that in football matches, teams that have more shots at goal also score more goals.

A football fan wanted to see if the journalist's claim was correct, so collected data for a random sample of 10 teams from the previous season. She collected, for each team, the number of shots at goal and the number of goals scored that season.

Her results are given in the table below.

Team	<i>A</i>	<i>B</i>	<i>C</i>	<i>D</i>	<i>E</i>	<i>F</i>	<i>G</i>	<i>H</i>	<i>I</i>	<i>J</i>
Number of shots at goal	902	1017	932	945	996	947	943	984	997	917
Number of goals scored	118	112	119	120	121	83	106	89	122	94

- (a) Calculate the value of Spearman's rank correlation coefficient between the number of shots at goal and the number of goals scored. Show your working clearly. (5)
- (b) Stating your hypotheses clearly, test whether the value that you calculated in part (a) supports the journalist's claim. Use a 5% level of significance. (4)

The football fan discovers that the number of shots at goal for team *A* is in fact 802 not 902

- (c) Without further calculations, state how this effects the answer to part (a). (1)



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[illegible]

5. A factory that manufactures bottles of juice receives complaints that its bottles contain less juice than is stated. The bottles are meant to contain 750 ml of juice.

An inspector visits the factory and takes a random sample of 6 bottles of juice. She measures the volumes of their contents, correct to the nearest half millilitre.

The results are as follows.

746.5 751.5 751.0 747.5 747.5 747.0

- (a) Stating your hypotheses clearly, test, at the 5% level of significance, whether or not there is evidence that the mean content of bottles is less than 750 ml.
You must make your working clear, showing the calculation of your test statistic and stating the critical value used.

(6)

- (b) State an assumption needed for this test.

(1)



DO NOT WRITE IN THIS AREA



6. A continuous random variable X has cumulative distribution function $F(x)$ given by

$$F(x) = \begin{cases} 0 & x < a \\ k(x - a)^2 & a \leq x \leq 2a \\ 1 & x > 2a \end{cases}$$

where k and a are constants.

- (a) Find k in terms of a . (2)

Given that $P(X > 2) = \frac{3}{4}$

- (b) explain why the value of $a = \frac{4}{3}$ (3)

- (c) Find the median of X , giving your answer to 3 significant figures. (3)

- (d) Without using a calculator, show that $E(X) = \frac{20}{9}$ (5)

- (e) Find the mode of X . (1)



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Question 6 continued

Lined area for writing the answer to Question 6.



Question 6 continued

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Question 6 continued

Lined area for writing answers.

(Total for Question 6 is 14 marks)



7. The diameter, X mm, of the circular mouth of bottles is normally distributed with mean 20 mm and standard deviation 0.1 mm. The diameter, Y mm, of the circular cross section of glass stoppers is normally distributed with mean 19.7 mm and standard deviation 0.1 mm. For a stopper to seal a bottle, the diameter of the mouth of the bottle minus the diameter of the stopper must be between 0 and 0.15 mm.

A bottle and a stopper are each selected at random.

- (a) Determine the probability that the stopper will seal the bottle.

(6)

Another bottle and a stopper are each selected at random.

- (b) Find the probability that the diameter of the mouth of the bottle is less than 19.5 mm smaller than twice the diameter of the stopper.

(6)



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Question 7 continued

Lined area for writing the answer to Question 7.



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Question 7 continued

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(Total for Question 7 is 12 marks)

TOTAL FOR PAPER IS 75 MARKS



Please check the examination details below before entering your candidate information

Candidate surname		Other names	
Pearson Edexcel		Centre Number	Candidate Number
Level 3 GCE			
Mock Paper Set 2			
(Time: 1 hour 30 minutes)		Paper Reference 9FM0/4B	
Further Mathematics			
Advanced			
Paper 4B: Further Statistics 2			
You must have: Mathematical Formulae and Statistical Tables, calculator			Total Marks

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- Answer the questions in the spaces provided
– *there may be more space than you need.*
- You should show sufficient working to make your methods clear. Answers without working may not gain full credit.
- Values from statistical tables should be quoted in full. If a calculator is used instead of the tables the value should be given to an equivalent degree of accuracy.
- Inexact answers should be given to three significant figures unless otherwise stated.

Information

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- There are 7 questions in this question paper. The total mark for this paper is 75.
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Advice

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Question 1 continued

Lined area for writing answers.

(Total for Question 1 is 8 marks)



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Question 2 continued

Lined area for writing the answer to Question 2.

(Total for Question 2 is 11 marks)



3. A village show has a jam making competition. The 8 jams submitted for the competition are tasted and each is awarded points by each of the three judges. Some of the points awarded are shown in Table 1

Jam	<i>A</i>	<i>B</i>	<i>C</i>	<i>D</i>	<i>E</i>	<i>F</i>	<i>G</i>	<i>H</i>
Judge 1	15	15	29	13	10	25	9	16
Judge 2	10	16	17	12	13	15	9	23
Judge 3		16	18	12		19	13	15

Table 1

- (a) Calculate Spearman's rank correlation coefficient between the points awarded by Judge 1 and the points awarded by Judge 2 (5)
- (b) Explain why Spearman's rank correlation coefficient is used in this case. (1)

Judge 3 had no tied ranks and only recorded his 6 highest points, which are shown in Table 1

Given that Spearman's rank correlation coefficient between the points awarded by

Judge 2 and the points awarded by Judge 3 is exactly $\frac{4}{7}$

- (c) explain which of jam *A* and jam *E* was awarded the lowest points from Judge 3. Show your working clearly. (5)



DO NOT WRITE IN THIS AREA

Question 3 continued

Lined area for writing the answer to Question 3.

(Total for Question 3 is 11 marks)



4. The continuous random variable X can only take values in the interval $[0, 1]$

After taking a series of observations of X , Adam suggests that the following probability density function might be a suitable model for X

$$f(x) = \begin{cases} kx(1-x^2) & 0 \leq x \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

- (a) Use algebraic integration to find the value of the constant k . (4)
- (b) Showing your working, find the mode of X using Adam's model. (2)

Beatrice takes an independent sample of observations of X and suggests that a more suitable model might be defined by the cumulative distribution function

$$H(x) = \begin{cases} 0 & x < 0 \\ \frac{\ln(1+x)}{\alpha} & 0 \leq x \leq 1 \\ 1 & x > 1 \end{cases}$$

- (c) Show that the constant $\alpha = \ln 2$ (2)
- (d) Using calculus and showing your working, find the exact value of the mean of X using Beatrice's model. (7)
- (e) Giving reasons for your answers, compare the skewness of the distributions suggested by Adam and Beatrice. (2)



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Question 4 continued

Lined area for writing the answer to Question 4.



Question 4 continued

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Question 4 continued

Lined area for writing answers.

(Total for Question 4 is 17 marks)



5. Two independent random variables X and Y have distributions $X \sim N(\mu_x, 3^2)$ and $Y \sim N(\mu_y, 4^2)$

A random sample of size n from X has mean $\bar{x} = 17$

A random sample of size $2n$ from Y has mean $\bar{y} = 23$

A test is carried out at the 5% significance level using the hypotheses

$$H_0 : \mu_y - \mu_x = 4 \quad H_1 : \mu_y - \mu_x > 4$$

Find the range of values of n for which H_0 is rejected.

(6)



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6. The random variables X and Y each have normal distributions with the same variance.

A researcher takes random samples from each distribution and calculates the following summary statistics,

$$\bar{x} = 76.5 \quad s_x^2 = 24$$

$$\bar{y} = 81.2 \quad s_y^2 = 27$$

Given that $n_y = n_x + 4$ and that the pooled estimate of variance, $s_p^2 = 25.8$

- (a) show that $n_x = 9$

(4)

The researcher believes that the population mean of Y , μ_y is greater than 2 more than the population mean of X , μ_x

- (b) Test whether or not the data provides evidence to support the researcher's belief at the 10% significance level. Show your working and state your hypotheses clearly.

(5)



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Question 6 continued

Lined area for writing the answer to Question 6.



Question 6 continued

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Question 6 continued

Lined area for writing answers.

(Total for Question 6 is 9 marks)



7. A shop supplies balloons for children's parties. Following complaints the manager tests the balloons to see at what pressures they burst. A random sample of 7 balloons is tested and the pressures at which they burst (x Pascals above atmospheric pressure) are given below

1734 1759 1777 1723 1747 1699 1740

The manager decides to use a t -distribution to calculate a 90% confidence interval for the mean pressure at which balloons burst.

- (a) State a necessary assumption that he needs to make in order for this calculation to be valid. (1)
- (b) Showing your working, use the data given above to calculate this confidence interval, giving your answers correct to 4 significant figures. (5)

The shop manager visits the balloon manufacturer. He is told that the standard deviation of the pressures at which the balloons burst is known to be 22 Pascals.

He is shown the results of an experiment in which 40 balloons were tested and produced a k % confidence interval for the mean pressure at which balloons burst of

$$1728.91 < \mu < 1741.09$$

- (c) Showing your working, calculate the value of k . (5)

The manufacturer repeated this experiment to produce 5 independent k % confidence intervals for the mean, each time using samples of size $n = 40$

- (d) Find the probability that all 5 of these confidence intervals contain the true value of the population mean. (2)



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Question 7 continued

Lined area for writing the answer to Question 7.



Question 7 continued

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(Total for Question 7 is 13 marks)

TOTAL FOR PAPER IS 75 MARKS



Write your name here

Surname

Other names

Pearson Edexcel
Level 3 GCE

Centre Number

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Candidate Number

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Further Mathematics

Advanced

Further Mathematics Option 2

Paper 4: Further Statistics 2

Sample Assessment Material for first teaching September 2017

Time: 1 hour 30 minutes

Paper Reference

9FM0/4E

You must have:

Mathematical Formulae and Statistical Tables, calculator

Total Marks

Candidates may use any calculator permitted by Pearson regulations. Calculators must not have the facility for algebraic manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

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Advice

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Answer ALL questions. Write your answers in the spaces provided.

1. The three independent random variables A , B and C each have a continuous uniform distribution over the interval $[0, 5]$.

(a) Find the probability that A , B and C are all greater than 3

(3)

The random variable Y represents the maximum value of A , B and C .

The cumulative distribution function of Y is

$$F(y) = \begin{cases} 0 & y < 0 \\ \frac{y^3}{125} & 0 \leq y \leq 5 \\ 1 & y > 5 \end{cases}$$

(b) Using algebraic integration, show that $\text{Var}(Y) = 0.9375$

(4)

(c) Find the mode of Y , giving a reason for your answer.

(2)

(d) Describe the skewness of the distribution of Y . Give a reason for your answer.

(1)

(e) Find the value of k such that $P(k < Y < 2k) = 0.189$

(3)

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Question 1 continued

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Question 1 continued

(Total for Question 1 is 13 marks)

2. A researcher claims that, at a river bend, the water gradually gets deeper as the distance from the inner bank increases. He measures the distance from the inner bank, b cm, and the depth of a river, s cm, at 7 positions. The results are shown in the table below.

Position	A	B	C	D	E	F	G
Distance from inner bank b cm	100	200	300	400	500	600	700
Depth s cm	60	75	85	76	110	120	104

The Spearman's rank correlation coefficient between b and s is $\frac{6}{7}$

- (a) Stating your hypotheses clearly, test whether or not the data provides support for the researcher's claim. Use a 1% level of significance.

(4)

- (b) Without re-calculating the correlation coefficient, explain how the Spearman's rank correlation coefficient would change if

(i) the depth for G is 109 instead of 104

(ii) an extra value H with distance from the inner bank of 800 cm and depth 130 cm is included.

(3)

The researcher decided to collect extra data and found that there were now many tied ranks.

- (c) Describe how you would find the correlation with many tied ranks.

(2)

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3. A nutritionist studied the levels of cholesterol, X mg/cm³, of male students at a large college. She assumed that X was distributed $N(\mu, \sigma^2)$ and examined a random sample of 25 male students. Using this sample she obtained unbiased estimates of μ and σ^2 as $\hat{\mu}$ and $\hat{\sigma}^2$

A 95% confidence interval for μ was found to be (1.128, 2.232)

- (a) Show that $\hat{\sigma}^2 = 1.79$ (correct to 3 significant figures)

(4)

- (b) Obtain a 95% confidence interval for σ^2

(3)

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4. The times, x seconds, taken by the competitors in the 100 m freestyle events at a school swimming gala are recorded. The following statistics are obtained from the data.

	No. of competitors	Sample mean $\bar{x}$	$\sum x^2$
Girls	8	83.1	55 746
Boys	7	88.9	56 130

Following the gala, a mother claims that girls are faster swimmers than boys. Assuming that the times taken by the competitors are two independent random samples from normal distributions,

- (a) test, at the 10% level of significance, whether or not the variances of the two distributions are the same. State your hypotheses clearly.

(7)

- (b) Stating your hypotheses clearly, test the mother's claim. Use a 5% level of significance.

(6)

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Question 4 continued

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Question 4 continued

(Total for Question 4 is 13 marks)

5. Scaffolding poles come in two sizes, long and short. The length L of a long pole has the normal distribution $N(19.6, 0.6^2)$. The length S of a short pole has the normal distribution $N(4.8, 0.3^2)$. The random variables L and S are independent.

A long pole and a short pole are selected at random.

- (a) Find the probability that the length of the long pole is more than 4 times the length of the short pole. Show your working clearly.

(6)

Four short poles are selected at random and placed end to end in a row. The random variable T represents the length of the row.

- (b) Find the distribution of T .

(3)

- (c) Find $P(|L - T| < 0.2)$

(4)

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Question 5 continued

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6. A random sample of 10 female pigs was taken. The number of piglets, x , born to each female pig and their average weight at birth, m kg, was recorded. The results were as follows:

Number of piglets, x	4	5	6	7	8	9	10	11	12	13
Average weight at birth, m kg	1.50	1.20	1.40	1.40	1.23	1.30	1.20	1.15	1.25	1.15

(You may use $S_{xx} = 82.5$ and $S_{mm} = 0.12756$ and $S_{xm} = -2.29$)

- (a) Find the equation of the regression line of m on x in the form $m = a + bx$ as a model for these results.

(2)

- (b) Show that the residual sum of squares (RSS) is 0.064 to 3 decimal places.

(2)

- (c) Calculate the residual values.

(2)

- (d) Write down the outlier.

(1)

- (e) (i) Comment on the validity of ignoring this outlier.

(ii) Ignoring the outlier, produce another model.

(iii) Use this model to estimate the average weight at birth if $x = 15$

(iv) Comment, giving a reason, on the reliability of your estimate.

(5)

Question 6 continued

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Question 6 continued

(Total for Question 6 is 12 marks)

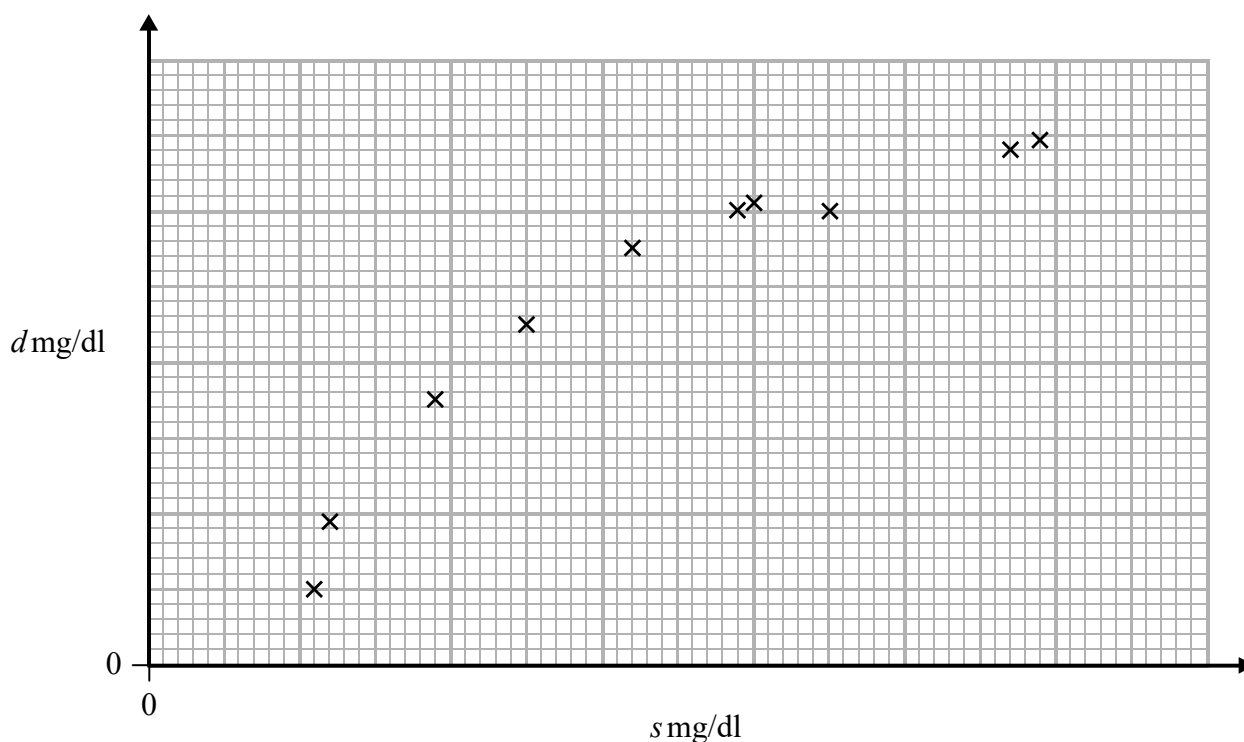
7. Over a period of time, researchers took 10 blood samples from one patient with a blood disease. For each sample, they measured the levels of serum magnesium, s mg/dl, in the blood and the corresponding level of the disease protein, d mg/dl. One of the researchers coded the data for each sample using $x = 10s$ and $y = 10(d - 9)$ but spilt ink over his work.

The following summary statistics and unfinished scatter diagram are the only remaining information.

$$\sum d^2 = 1081.74 \quad S_{ds} = 59.524$$

and

$$\sum y = 64 \quad S_{xx} = 2658.9$$



- (a) Use the formula for S_{xx} to show that $S_{ss} = 26.589$ (3)
- (b) Find the value of the product moment correlation coefficient between s and d . (4)
- (c) With reference to the unfinished scatter diagram, comment on your result in part (b). (1)

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Question 7 continued

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TOTAL FOR PAPER IS 75 MARKS

Please check the examination details below before entering your candidate information

Candidate surname

Other names

**Pearson Edexcel
Level 3 GCE**

Centre Number

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Candidate Number

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Specimen Paper

(Time: 1 hour 30 minutes)

Paper Reference **9FM0/4B**

**Further Mathematics
Advanced
Paper 4B: Further Statistics 2**

You must have:

Mathematical Formulae and Statistical Tables, calculator

Total Marks

**Candidates may use any calculator permitted by Pearson regulations.
Calculators must not have the facility for algebraic manipulation,
differentiation and integration, or have retrievable mathematical
formulae stored in them.**

Instructions

- Use **black** ink or ball-point pen.
- If pencil is used for diagrams/sketches/graphs it must be dark (HB or B).
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- Answer **all** questions and ensure that your answers to parts of questions are clearly labelled.
- Answer the questions in the spaces provided
– *there may be more space than you need.*
- You should show sufficient working to make your methods clear.
Answers without working may not gain full credit.
- Answers should be given to three significant figures unless otherwise stated.

Information

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- There are 7 questions in this question paper. The total mark for this paper is 75.
- The marks for **each** question are shown in brackets
– *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ►

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- The editor then counted the number of words in each of the 10 articles and her results are given in the table.

Article	<i>A</i>	<i>B</i>	<i>C</i>	<i>D</i>	<i>E</i>	<i>F</i>	<i>G</i>	<i>H</i>	<i>I</i>	<i>J</i>
Read rank	4	6	1	10	8	2	7	5	3	9
No. of words	567	2613	350	2580	1214	485	785	819	620	726

- (a) Calculate Spearman's rank correlation coefficient for these data. (4)
- (b) Test the editor's belief. Use a 5% significance level and state your hypotheses clearly. (3)

Question 1 continued

Lined area for writing the answer to Question 1.

(Total for Question 1 is 7 marks)



2. New recruits to *Sonarep* are given a two-part test covering spatial and numerical skills. The scores on each part of the test are normally distributed. The recruitment manager believes that the mean score for spatial skills is more than the mean score for numerical skills and wishes to find a 95% confidence interval for the mean difference. She took a random sample of 8 recruits and recorded their scores on each part of the test. The results are given in the table.

Recruit	<i>A</i>	<i>B</i>	<i>C</i>	<i>D</i>	<i>E</i>	<i>F</i>	<i>G</i>	<i>H</i>
Spatial skills	41	38	29	45	37	48	33	36
Numerical skills	35	32	31	37	38	36	33	26

- (a) Calculate a 95% confidence interval for the difference in mean scores between the two parts of the test. (5)
- (b) State, giving a reason, whether or not the confidence interval supports the recruitment manager's belief. (2)

The deputy manager treated these data as two independent samples and obtained a 95% confidence interval for the difference in mean scores of $(-0.67, 10.42)$

- (c) Explain why this treatment of these data is not correct. (1)



Question 2 continued

Handwriting practice area with 30 horizontal lines.

(Total for Question 2 is 8 marks)



3. George took a random sample of 8 apples from the tree in his garden. The weight, x grams, of each apple was recorded. The results are summarised by

$$\sum x = 752 \text{ and } \sum x^2 = 73\,094$$

- (a) Calculate a 90% confidence interval for the standard deviation, σ , of the weights of apples from the tree in George's garden.

(6)

- (b) State an assumption about the weights of apples from the tree that is needed to carry out the calculation in part (a).

(1)

- (c) State, giving a reason, whether or not the confidence interval in part (a) supports the assertion that $\sigma = 26$ grams.

(2)

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Question 3 continued

Handwriting practice area with 28 horizontal lines.

(Total for Question 3 is 9 marks)



4. As part of a winter sports day both boys and girls complete a cross country course. The times taken, in minutes, by a random sample of 9 girls are given below.

35.6 28.7 23.4 42.1 38.9 31.6 34.7 26.3 37.2

- (a) Find unbiased estimates of the mean and variance of the times taken by the girls to complete the course.

(2)

The times, x minutes, taken by a random sample of 6 boys to complete the course are summarised by the following statistics

$$\bar{x} = 27.45 \quad s_x^2 = 15.77$$

- (b) Test, at the 10% level of significance, whether or not there is a difference in the variances of the times taken by the girls and the times taken by the boys to complete the course. State your hypotheses clearly.

(4)

A sports teacher at the school believes that the mean time taken by the girls to complete the course is more than 2 minutes longer than the mean time taken by the boys.

- (c) Stating your hypotheses clearly, test the teacher's belief at the 5% level of significance.

(6)

- (d) State an assumption about the distribution of the times taken to complete the course for the tests in part (b) and part (c) to be valid.

(1)

- (e) Explain the relevance of the test in part (b) to the test in part (c).

(1)

[illegible]

Question 4 continued

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Question 4 continued

Lined area for writing the answer to Question 4.

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Question 4 continued

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(Total for Question 4 is 14 marks)



5. *Eusain Engineering* manufactures bolts in large batches. The lengths of the bolts, in mm, follow a normal distribution with mean 38 mm and standard deviation 0.3 mm.

A random sample of 3 bolts is taken and the lengths measured.

Find the probability that the difference between the mean length of the first 2 bolts and the mean length of all 3 bolts is greater than 0.2 mm.

(7)

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Question 5 continued

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Question 5 continued

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Question 5 continued

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(Total for Question 5 is 7 marks)



6. The continuous random variable X has probability density function

$$f(x) = \begin{cases} \frac{1}{2\pi}(1 - \cos x) & 0 \leq x \leq 2\pi \\ 0 & \text{otherwise} \end{cases}$$

- (a) Sketch $f(x)$. (2)
- (b) Use algebraic integration to show that $E(X^2) = \frac{4\pi^2}{3} - 2$ (4)
- (c) Hence find the standard deviation of X . (2)

The lower quartile of X is 2.31 (correct to 3 significant figures).
Jeff suggests using the normal distribution, with the same mean and standard deviation as X , to estimate the interquartile range of X .

- (d) Find the percentage error of this estimate. (4)



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Question 6 continued

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Question 6 continued

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Question 6 continued

Lined area for writing answers.

(Total for Question 6 is 12 marks)



7. A local chamber of commerce researcher collected information from a random sample of 8 new retail companies. The number of employees, m , and the company turnover in £10 000, t , was recorded for each company. The results are summarised below

$$\bar{m} = 8 \quad \bar{t} = 62 \quad S_{mm} = 246 \quad S_{tt} = 20461.16 \quad S_{mt} = 2076.8$$

- (a) Calculate the product moment correlation coefficient for these data. (2)
- (b) Stating your hypotheses clearly, test at the 5% level of significance whether or not there is a positive correlation between number of employees and company turnover. (3)

Steve owns a small retail company and is thinking of increasing the number of people he employs. He believes that there may be a relationship between number of employees and company turnover and he wants to use this to estimate the change in turnover.

- (c) (i) State, giving a reason, the form of a suitable relationship Steve can use that is consistent with your answer to part (b). (1)
- (ii) Find the equation of this model. (3)
- (d) Use this model to estimate the change in turnover, in £, if Steve increases the number of employees from 13 to 16 (2)

Question 7 continues on page 22



Question 7 continues on page 22



Question 7 continued

The residuals for the model Steve used are given in the table below.

Number of employees	1	2	4	6	9	10	14	18
Residual	26.6	13.5	-8.8	-14.5	-21.9	-26.4	9.8	21.9

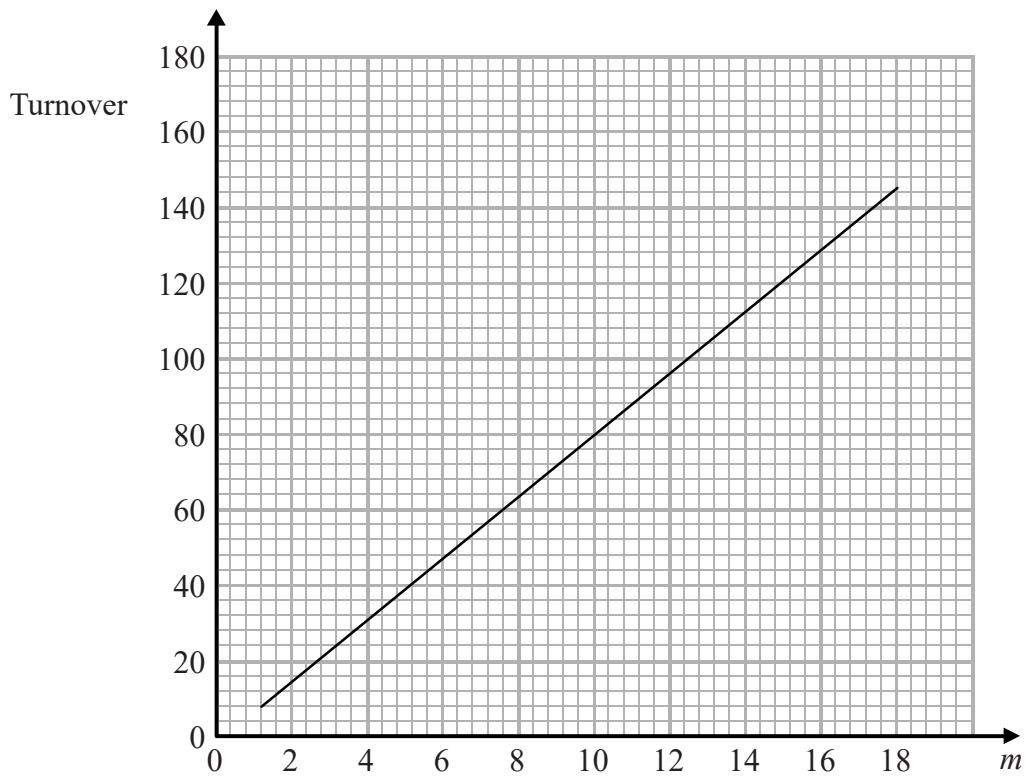


Figure 1

Figure 1 shows a sketch of Steve's model.

- (e) On Figure 1 sketch the scatter diagram for Steve's data. (2)
- (f) Use your scatter diagram and the table of residuals to refine the model and calculate a revised estimate for your answer to part (d). Explain clearly how you calculated your estimate. (2)
- Given that there are 4 positive and 4 negative residuals,
- (g) find the probability that a random pattern of residuals is $++-- --++$. (2)
- With reference to your answers to part (a) and part (g),
- (h) suggest what Steve should do to investigate further. (1)

Turn to page 24 for a spare grid if you need to redraw your scatter diagram.



Question 7 continued

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(TOTAL FOR QUESTION 7 IS 10 MARKS)

24

